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Contents Table

Functions

- Vertical Line Test
- $|x|$
  - Basic Definition of Absolute Value: $|x| = \begin{cases} x, & x \geq 0 \\ -x, & x < 0 \end{cases}$
    - $\sqrt{x^2} = |x|$
    - $e^{\frac{-|x|}{2}}$ (graph this)
- Anton – Chapter 0 – Page 8: "The Effect of Algebraic Operations on the Domain"

Limits

1. Easy ones
   - **Direct Substitution**

Direct Substitution – Theorem 1.4.10

If $f(x)$ is a polynomial or rational function, and a is in the domain of f, then:

$$\lim_{x \to a} f(x) = f(a).$$

© Week1 Slides, Page 154
@ CLP-1 by Joel, Andrew, Elyse

$$\lim_{x \to 5}(x^2 - 4x + 3) \qquad \lim_{x \to 1}\left(x^7 - 2x^5 + 1\right)^{35} \qquad \lim_{x \to 2} x(x-1)(x+1) \qquad \text{(a)} \lim_{x \to 3}\left(\frac{x^2 - 9}{x + 3}\right) \qquad \lim_{x \to 0}\left(\frac{6x - 9}{x^3 - 12x + 3}\right)$$

   - **Factorising**

$$\lim_{x \to 2}\left(\frac{x - 2}{x^2 - 4}\right) \qquad \lim_{x \to 4}\frac{x^2 - 4x}{x^2 - 16} \qquad \lim_{x \to 1}\frac{x^3 - x^2}{x - 1} \qquad \lim_{x \to 2}\frac{x^2 + x - 6}{x - 2} \qquad \text{(b)} \lim_{x \to -3}\left(\frac{x^2 - 9}{x + 3}\right)$$

     - Anton – Chapter 1.2 – Exercises 3-14
     - Anton – Chapter 1.2 – Exercise 40 [This is how exam questions should be]

   - Look out for questions that look like "Limits @ Infinity"

$$\lim_{x \to 2}\frac{x^2 - 4x + 4}{x^2 + x - 6} \qquad \lim_{x \to 2}\frac{2x^2 - 5x + 2}{5x^2 - 7x - 6}$$

2. **Rationalizing the Denominator**

$$\lim_{x \to 9}\frac{x - 9}{\sqrt{x} - 3} \qquad \lim_{y \to 4}\frac{4 - y}{2 - \sqrt{y}} \qquad \lim_{x \to 0}\frac{x}{\sqrt{x + 1} - 1}$$

3. **Conjugate**
   - Anton – Chapter 1.2 – Exercises 37,38

$$\lim_{x \to 0}\frac{\sqrt{x + 4} - 2}{x} \qquad \lim_{x \to 0}\frac{\sqrt{x^2 + 4} - 2}{x}$$

4. **1 Sided Limits**
   - PieceWise Functions
   - Anton – Chapter 1.2 – Exercises 15-28

$$\lim_{x \to 0}\frac{x}{|x|} \qquad \lim_{x \to 0}\frac{1}{|2 - x|} \qquad \lim_{x \to 3}\frac{x}{x - 3} \qquad \lim_{x \to 2}\frac{x}{x^2 - 4} \qquad \text{(c)} \lim_{x \to 6}\left(\frac{x + 6}{x^2 - 36}\right) \qquad \lim_{x \to 4}\frac{3 - x}{x^2 - 2x - 8}$$

   - Notice the difference between (a), (b), (c)

5. $\lim\limits_{x\to\infty}$

- Type I

$$\lim_{x\to+\infty}\left(\frac{3x+5}{6x-8}\right)=\lim_{x\to+\infty}\frac{3x}{6x}=\frac{3}{6}$$

$$\lim_{x\to+\infty}\frac{2x^2-5x+2}{5x^2-7x-6}=\lim_{x\to+\infty}\frac{2x^2}{5x^2}=\frac{2}{5}$$

$$\lim_{x\to+\infty}\left(\frac{6-x^3}{7t^3+3}\right)=\lim_{x\to+\infty}\frac{-x^3}{7x^3}=\frac{-1}{7}$$

$$\lim_{x\to-\infty}\frac{x+4x^3}{1-x^2+7x^3}=\lim_{x\to-\infty}\frac{4x^3}{7x^3}=\frac{-4}{-7}$$

$$\lim_{x\to+\infty}\sqrt[3]{\frac{3x+5}{6x-8}}=\sqrt[3]{\lim_{x\to+\infty}\frac{3x}{6x}}=\sqrt[3]{\frac{3}{6}}$$

$$\lim_{x\to+\infty}\sqrt[3]{\frac{3x^7-4x^5}{2x^7+1}}=\sqrt[3]{\lim_{x\to+\infty}\frac{3x^7}{2x^7}}=\sqrt[3]{\frac{3}{2}}$$

- Type II

$$\lim_{x\to+\infty}\frac{4x^2-x}{2x^3-5}=\lim_{x\to+\infty}\frac{4x^2}{2x^3}=\lim_{x\to+\infty}\frac{4}{2x}=\lim_{x\to+\infty}\frac{1}{2x}=\lim_{x\to+\infty}\frac{1}{x}=\frac{1}{\infty}=0^+$$

$$\lim_{x\to-\infty}\frac{3}{x+4}=\lim_{x\to-\infty}\frac{1}{x}=\frac{1}{-\infty}=0^- \qquad \Longrightarrow \text{approaching 0, yes, but from the bottom-half of y-axis}$$

$$\lim_{x\to+\infty}\frac{1}{x-12}=\lim_{x\to+\infty}\frac{1}{x}=\frac{1}{+\infty}=0^+$$

$$\lim_{x\to-\infty}\frac{x-2}{x^2+2x+1}=\lim_{x\to-\infty}\frac{x}{x^2}=\lim_{x\to-\infty}\frac{1}{x}$$

- SideNote: $\lim\limits_{-\infty}$

$$\lim_{x\to-\infty}\frac{x}{x^2}=\lim_{x\to-\infty}\frac{1}{x}\cdot\frac{\cancel{x}}{\cancel{x}} \qquad \Longrightarrow \lim_{x\to-\infty}\frac{1}{x}\cdot\frac{\text{-ve}}{\text{-ve}} \qquad \Longrightarrow \qquad \lim_{x\to-\infty}\frac{1}{x}\cdot 1=\lim_{x\to-\infty}\frac{1}{x}=\frac{1}{-\infty}=0^-$$

- Type III

$$\lim_{x\to+\infty}\frac{7-6x^5}{x+3}=\lim_{x\to+\infty}\frac{-6x^5}{x}=\lim_{x\to+\infty}(-6)\cdot\frac{x^4}{1}=(-6)\cdot\frac{(+\infty)^4}{1}=(-6)\cdot\frac{\infty}{1}=-\infty$$

$$\lim_{x\to-\infty}\frac{7-6x^5}{x+3}=\lim_{x\to-\infty}\frac{-6x^5}{x}=\lim_{x\to-\infty}(-6)\cdot\frac{x^4}{1}=(-6)\cdot\frac{(-\infty)^4}{1}=(-6)\cdot\frac{\infty}{1}=-\infty$$

$$\lim_{x\to-\infty}\frac{7-6x^4}{x+3}=\lim_{x\to-\infty}\frac{-6x^4}{x}=\lim_{x\to-\infty}(-6)\cdot\frac{x^3}{1}=(-6)\cdot\frac{(-\infty)^3}{1}=(-6)\cdot\frac{-\infty}{1}=+\infty$$

- Anton – Chapter 1.3 – Page 93: "A Quick method for finding limits of ..."
- Anton – Chapter 1.3 – Exercises 09-24
- Exercises

$$\lim_{x\to-\infty}\frac{3}{x+4} \qquad \lim_{x\to+\infty}\frac{1}{x-12} \qquad \lim_{x\to-\infty}\frac{x-2}{x^2+2x+1} \qquad \lim_{x\to+\infty}\frac{6-x^3}{7x^3+3} \qquad \lim_{x\to-\infty}\frac{x+4x^3}{1-x^2+7x^3}$$

5. $\lim\limits_{x\to\infty}$

- Type IV – Divide by $|x|$
  - Anton – Chapter 1.3 – Page 94: "Limits Involving Radicals"

$$\lim_{x\to +\infty} \frac{\sqrt{x^2+2}}{3x-6} = \lim_{x\to +\infty} \frac{\left(\frac{\sqrt{x^2+2}}{|x|}\right)}{\left(\frac{3x-6}{|x|}\right)} = \lim_{x\to +\infty} \frac{\left(\frac{\sqrt{x^2+2}}{\sqrt{x^2}}\right)}{\left(\frac{3x-6}{+x}\right)} \qquad \because |x| = \sqrt{x^2} \quad \text{and} \quad \because |x| = \begin{cases} x, & x \geq 0 \\ -x, & x < 0 \end{cases}$$

$$\lim_{x\to -\infty} \frac{\sqrt{x^2+2}}{3x-6} = \lim_{x\to -\infty} \frac{\left(\frac{\sqrt{x^2+2}}{|x|}\right)}{\left(\frac{3x-6}{|x|}\right)} = \lim_{x\to -\infty} \frac{\left(\frac{\sqrt{x^2+2}}{\sqrt{x^2}}\right)}{\left(\frac{3x-6}{-x}\right)} \qquad \because |x| = \sqrt{x^2} \quad \text{and} \quad \because |x| = \begin{cases} x, & x \geq 0 \\ -x, & x < 0 \end{cases}$$

  - Anton – Chapter 1.3 – Exercises 25-30
  - Quick Method

$$\lim_{x\to -\infty} \frac{\sqrt{5x^2-2}}{x+3} = \lim_{x\to -\infty} \frac{\sqrt{5x^2}}{x} = \frac{\sqrt{5}}{-1} \qquad \lim_{x\to +\infty} \frac{\sqrt{5x^2-2}}{x+3} = \lim_{x\to +\infty} \frac{\sqrt{5x^2}}{x} = \frac{\sqrt{5}}{+1}$$

$$\lim_{x\to -\infty} \frac{2-x}{\sqrt{7+6x^2}} = \lim_{x\to -\infty} \frac{-x}{\sqrt{6x^2}} = \frac{-(-1)}{\sqrt{6}}$$

$$\lim_{x\to -\infty} \frac{\sqrt{3x^4+x}}{x^2-8} = \lim_{x\to -\infty} \frac{\sqrt{3x^4}}{x^2} = \frac{\sqrt{3}}{1}$$

- Type V: Multiply by Conjugate

$$\lim_{x\to +\infty} \sqrt{x^6+5} - x^3 \qquad \lim_{x\to +\infty} \sqrt{x^6+5x^3} - x^3$$

- Anton – Chapter 1.3 – Exercises 31-32 [Conjugate]

$$\lim_{x\to +\infty} \sqrt{x^2+3} - x \qquad \lim_{x\to +\infty} \sqrt{x^2-3x} - x$$

- Type VI
  - Anton – Chapter 1.3 – Page 95: "End behavior of Trigonometric, Exponential, and Logarithmic Funcs"
  - Anton – Chapter 1.3 – Exercises 33-36 – Divide by e^x

$$\lim_{x\to -\infty} \frac{1-e^x}{1+e^x}$$

  - Anton – Chapter 1.3 – Exercises 37-38 – $\ln()$

$$\lim_{x\to +\infty} \ln\left(\frac{2}{x^2}\right) \qquad \lim_{x\to 0^+} \ln\left(\frac{2}{x^2}\right)$$

6. **Squeeze Theorem / SandWich Theorem**

$$\lim_{h\to 0} \frac{\sin(h)}{h} = 1$$

$$\lim_{h\to 0} \cos\left(\frac{1}{h}\right) = \text{range}[-1,\, 1] = \text{D.N.E.}$$

$$\lim_{h\to 0} h \cdot \cos\left(\frac{1}{h}\right) = 0$$

$$\lim_{h\to 0} h \cdot \sin\left(\frac{1}{h}\right) = 0$$

$$\begin{aligned}
\lim_{h\to 0} \frac{\cos(h)-1}{h} &= \lim_{h\to 0} \frac{\cos(h)\ -1}{h} \cdot \frac{\cos(h)+1}{\cos(h)+1} \\
&= \lim_{h\to 0} \frac{\cos^2(h)-1^2}{h} \cdot \frac{1}{\cos(h)+1} \\
&= \lim_{h\to 0} \frac{1-\sin^2(h)-1^2}{h} \cdot \frac{1}{\cos(h)+1} \\
&= \lim_{h\to 0} \frac{-\sin^2(h)}{h} \cdot \frac{1}{\cos(h)+1} \\
&= \lim_{h\to 0} \frac{\sin(h)}{h} \cdot -\frac{\sin(h)}{\cos(h)+1}
\end{aligned}$$